WORKED SOLUTION

STEP II 2019, Question 12

Probability and Statistics • Continuous Distributions and Quantiles

STEP II 2019, Question 12

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

The cumulative distribution function here is simply x^n on $[0,1]$, so every quantile can be written down exactly: the value with a fraction p of the distribution below it is $p^{1/n}$. Comparing the mean with a quantile then becomes a comparison of powers, and raising both sides to the nth power turns it into a statement about $\left(1+\frac{1}{n}\right)^n$, which is exactly what the binomial expansion is built to estimate.

Solution

Part (i)

Integrating against the density,

$$\mu = \mathrm{E}(X) = \int_0^1 x \cdot nx^{n-1}\,\mathrm{d}x = \int_0^1 nx^n\,\mathrm{d}x = \frac{n}{n+1},$$
$$\mathrm{E}(X^2) = \int_0^1 nx^{n+1}\,\mathrm{d}x = \frac{n}{n+2}.$$

Hence

$$\sigma^2 = \mathrm{E}(X^2) - \mu^2 = \frac{n}{n+2} - \frac{n^2}{(n+1)^2} = \frac{n\left[(n+1)^2 - n(n+2)\right]}{(n+1)^2(n+2)}$$
$$= \frac{n\left[n^2+2n+1-n^2-2n\right]}{(n+1)^2(n+2)} = \frac{n}{(n+1)^2(n+2)},$$

as required.

Part (ii)

For $0 \le x \le 1$ the distribution function is

$$\mathrm{F}(x) = \mathrm{P}(X \le x) = \int_0^x nt^{n-1}\,\mathrm{d}t = x^n.$$

With $n = 2$, $\mathrm{F}(x) = x^2$. The lower quartile q_1 satisfies $q_1^2 = \frac{1}{4}$ and the upper quartile q_3 satisfies $q_3^2 = \frac{3}{4}$; both lie in $[0, 1]$, so we take positive roots:

$$q_1 = \tfrac{1}{2}, \qquad q_3 = \tfrac{\sqrt{3}}{2}, \qquad q_3 - q_1 = \frac{\sqrt{3}-1}{2}.$$

From part (i), $\sigma^2 = \dfrac{2}{9 \cdot 4} = \dfrac{1}{18}$, so $\sigma = \dfrac{1}{3\sqrt{2}} = \dfrac{\sqrt{2}}{6}$ and $2\sigma = \dfrac{\sqrt{2}}{3}$.

We want to show $\dfrac{\sqrt{3}-1}{2} < \dfrac{\sqrt{2}}{3}$. Multiplying by 6, this is equivalent to

$$3\sqrt{3} - 3 < 2\sqrt{2} \iff 3\sqrt{3} < 3 + 2\sqrt{2}.$$

Both sides of the last inequality are positive, so it is equivalent to the inequality between their squares:

$$27 < 9 + 12\sqrt{2} + 8 = 17 + 12\sqrt{2} \iff 10 < 12\sqrt{2},$$

which is true because $\sqrt{2} > 1$ gives $12\sqrt{2} > 12 > 10$. Every step was an equivalence, so working back up the chain, the interquartile range is less than 2σ.

Ask yourself Why is it safe to square both sides here, but not in general?

Squaring preserves an inequality $a < b$ only when a and b are non-negative (for example $-3 < 2$ but $9 > 4$). That is why we first rearranged to $3\sqrt{3} < 3 + 2\sqrt{2}$, where both sides are clearly positive, rather than squaring $3\sqrt{3} - 3 < 2\sqrt{2}$ with a subtraction still inside.

Part (iii)

By the binomial theorem,

$$(1 + x)^n = 1 + nx + \frac{n(n-1)}{2}\, x^2 + \cdots,$$

and the $(k+1)$th term is

$$\binom{n}{k} x^k = \frac{n(n-1)\cdots(n-k+1)}{k!}\, x^k.$$

Median and lower quartile. Since $\mathrm{F}(x) = x^n$, the median m satisfies $m^n = \frac{1}{2}$ and the lower quartile q satisfies $q^n = \frac{1}{4}$, so

$$m = \left(\tfrac{1}{2}\right)^{1/n}, \qquad q = \left(\tfrac{1}{4}\right)^{1/n}.$$

All of μ, m, q are positive, and $t \mapsto t^n$ is strictly increasing for $t > 0$, so we may compare them by comparing their nth powers. Also

$$\mu^n = \left(\frac{n}{n+1}\right)^n = \frac{1}{\left(1 + \frac{1}{n}\right)^n}.$$

Write $A = \left(1+\frac{1}{n}\right)^n$. Then

$$\mu < m \iff \mu^n < \tfrac{1}{2} \iff A > 2, \qquad \mu > q \iff \mu^n > \tfrac{1}{4} \iff A < 4.$$

μ is less than the median. Put $x = \frac{1}{n}$ in the expansion. Every term is non-negative, so A is at least the sum of its first three terms:

$$A \geq 1 + n\cdot\frac{1}{n} + \frac{n(n-1)}{2}\cdot\frac{1}{n^2} = 2 + \frac{n-1}{2n} > 2,$$

where the last inequality is strict because $n > 1$. (For $n \geq 2$ the third term really is present in the expansion.) Hence $\mu < m$.

μ is greater than the lower quartile. With $x = \frac{1}{n}$, the $(k+1)$th term is

$$\binom{n}{k}\frac{1}{n^k} = \frac{1}{k!}\cdot\frac{n}{n}\cdot\frac{n-1}{n}\cdots\frac{n-k+1}{n} \leq \frac{1}{k!},$$

since each of the k fractions $\frac{n-j}{n}$ $(0 \leq j \leq k-1)$ lies between 0 and 1. (For $k = 0$ the term is $1 = \frac{1}{0!}$.) Summing over $k = 0, 1, \ldots, n$,

$$A \leq 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!} < 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots < 4,$$

the middle inequality holding because the infinite series contains further positive terms, and the last being the result we were allowed to assume. Hence $A < 4$, so $\mu > q$.

Putting the two together, the lower quartile is less than μ, which is less than the median, for every integer $n > 1$.

Ask yourself Why does comparing nth powers settle the comparison of μ with m?

For positive numbers a and b, $a < b$ if and only if $a^n < b^n$, because t^n is strictly increasing on $t > 0$. Our three numbers are all positive, so nothing is lost by raising to the nth power, and doing so removes the awkward nth roots in m and q.

Checks

- For $n = 2$: $\mu = \frac{2}{3}$, and $\sigma^2 = \frac{2}{9\cdot4} = \frac{1}{18}$, which matches $\mathrm{E}(X^2) - \mu^2 = \frac{1}{2} - \frac{4}{9}$.
- Numerically, for $n = 2$ the interquartile range is 0.366 and $2\sigma = 0.471$.
- For $n = 2$ the three values are $q = 0.5$, $\mu \approx 0.667$, $m \approx 0.707$; for $n = 10$ they are $q \approx 0.871$, $\mu \approx 0.909$, $m \approx 0.933$. The ordering $q < \mu < m$ was confirmed by computer for every n from 2 to 2000.
- As $n \to \infty$, $A = \left(1+\frac{1}{n}\right)^n \to \mathrm{e} \approx 2.718$, which sits comfortably between 2 and 4.

Common slips

- Squaring $3\sqrt{3} - 3 < 2\sqrt{2}$ directly, without first checking that both sides are positive.
- Getting the direction wrong when taking reciprocals: $\mu^n < \frac{1}{2}$ corresponds to $A > 2$, not $A < 2$.
- Claiming the $(k+1)$th term equals $\frac{1}{k!}$ rather than being at most $\frac{1}{k!}$.

Beyond STEP

The quantity $\left(1 + \frac{1}{n}\right)^n$ increases with n to the limit e, and the bound you proved, term by term against $\sum 1/k!$, is the first step in showing that $\mathrm{e} = \sum_{k \geq 0} 1/k!$, a standard result in a first analysis course. The distribution itself is a Beta$(n, 1)$ distribution; it is also the distribution of the maximum of n independent uniform random variables on $[0, 1]$, since $\mathrm{P}(\max \leq x) = x^n$. For more on both threads, see Blitzstein and Hwang, *Introduction to Probability*, and Abbott, *Understanding Analysis*.

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